Technical Supplement: "It's the Elicitation, Not the Eyes: Diagnosing VLM Failures in Slide Quality Inspection and Routing a Symbolic–Neural Critic"

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This supplement extends the main paper with material referenced from the body but omitted there for space: the SlideAudit crosswalk and full transfer table (§1), the coverage precision companion (§2), the full multi-reward / VLM-judge audit grid with per-class preference accuracies (§3), the full page-semantic examiner table including the deck-scope negative result (§4), and the extended figures (§6). All numbers are transcribed verbatim from the frozen analysis reports and rollout records that accompany this submission as the Code and Data Supplement; the held-out and stress-test addenda are identified explicitly below.

1 SlideAudit crosswalk

Table 1 gives the bidirectional map between our G/S defect classes and the 19-dimensional expert-derived SLIDEAUDIT taxonomy, referenced from §3 of the main paper. Empty entries are intentional off-taxonomy semantic or deck-level defects; G7 refines, rather than duplicates, SLIDEAUDIT's content-overflow/cut-off category by adding the declared-legal-box criterion that makes the defect linter-blind.

Table 1: Crosswalk between our G/S classes and the 19-dimensional SLIDEAUDIT taxonomy.

Our class	SLIDEAUDIT class	Relation
G1 text overflow	Content Overflow/Cut-off	equivalent
G2 element overlap	Occluded Content	equivalent
G3 alignment offset	Content Alignment Issues	equivalent
G4 font-size inconsistency	Improper Font Sizing	equivalent
G5 brand-color violation	Excessive or Inconsistent Color Usage; Inappropriate or Mismatched Color Combinations	equivalent
G6 margin violation	Unbalanced Space Distribution	near-equivalent; safe-margin bleed is a special case
G7 render-containment overflow	Content Overflow/Cut-off; related to Improper Image Sizing	our extension; same visual symptom plus declared-legal-box criterion
S1 title/body mismatch	–	off-taxonomy content semantics
S2 narrative-order break	–	off-taxonomy deck-level ordering
S3 terminology inconsistency	–	off-taxonomy cross-slide terminology
S4 density violation	Excessive Text Volume	equivalent
S5 missing logic section	–	off-taxonomy deck-level completeness
S6 image/text contradiction	Irrelevant Visual Content (nearest)	nearest only; contradiction rather than relevance

1.1 Per-class image-only transfer results

Table 2 reports every mapped class evaluated on SlideAudit. This is a third-party, image-only dataset: no element IR is available, so the symbolic linter cannot run and the hybrid reduces to its VLM branch. We pair labelled-present examples with confidently absent controls and report named-attribution balanced accuracy and precision, both with 95% Wilson intervals. The slight S4 C0 count difference (39 rather than 40 positives) is the valid-record count after parsing.

Table 2: Complete SlideAudit transfer results for Qwen3.5-27B, image-only modality. Each result is balanced accuracy [95% Wilson]; precision [95% Wilson]. n_+/n_- gives the effective scored counts.

class	C0 whole-taxonomy	C3 atomic	n_+/n_-
G1 overflow	.550 [.476,.615]; 1.000 [.510,1]	.625 [.501,.727]; .812 [.570,.934]	40/40
G2 overlap	.963 [.857,.987]; .930 [.814,.976]	.938 [.818,.980]; .949 [.831,.986]	40/40
G3 alignment	.558 [.439,.687]; .571 [.250,.842]	.723 [.552,.850]; .706 [.469,.867]	21/40
G4 font size	.588 [.479,.681]; .818 [.523,.949]	.637 [.506,.745]; .789 [.567,.915]	40/40
G5 colour	.516 [.459,.581]; 1.000 [.207,1]	.814 [.668,.900]; .913 [.732,.976]	31/40
G6 margin	.512 [.458,.564]; 1.000 [.207,1]	.550 [.431,.660]; .643 [.388,.837]	40/40
S4 density	.680 [.553,.776]; .889 [.672,.969]	.738 [.600,.834]; .880 [.700,.958]	39/40; 40/40

1.2 Exact evaluated prompt templates

The following are the complete textual templates used by the two conditions; the image occupies the indicated image part. Bracketed fields are populated from each manifest record. C0 receives all nine page candidates in one call.

C0 system

```
You are a slide quality examiner.  Inspect the given single slide page for the requested
page-scoped defect types.  Output ONLY one valid JSON object matching PageExamResult.  No prose, no
code fences.
```

C0 user

```
[IMAGE]
PAGE_ID: [page_id]
render=[image_width_px]x[image_height_px] scale=([scale_x],[scale_y]) dpi=[dpi] renderer=[renderer]
SCENE: [scene]
TEMPLATE_ID: [template_id]
PAGE_INDEX: [page_index]
TASK_BRIEF: [task_brief]
CHECK_SCOPE: ['G1_TEXT_OVERFLOW', 'G2_ELEMENT_OVERLAP', 'G3_ALIGNMENT_OFFSET',
'G4_FONT_SIZE_INCONSISTENCY', 'G5_BRAND_COLOR_VIOLATION', 'G6_MARGIN_VIOLATION',
'S1_TITLE_BODY_MISMATCH', 'S4_DENSITY_RULE_VIOLATION', 'S6_IMAGE_TEXT_CONTRADICTION']
TASK: Decide for each candidate type whether the defect is PRESENT.
Consider ONLY these candidate defect types:
G1_TEXT_OVERFLOW: text overflows or spills beyond the bounds of its box/placeholder;
G2_ELEMENT_OVERLAP: two elements visually overlap or collide when they should not;
G3_ALIGNMENT_OFFSET: an element is misaligned / offset from where it should sit;
G4_FONT_SIZE_INCONSISTENCY: inconsistent font sizes among text that should match;
G5_BRAND_COLOR_VIOLATION: a color is off-brand / wrong for an element;
G6_MARGIN_VIOLATION: an element bleeds past the page margin or off the canvas edge;
S1_TITLE_BODY_MISMATCH: the slide title does not match the body content / topic;
S4_DENSITY_RULE_VIOLATION: the slide is overcrowded / too dense with text;
S6_IMAGE_TEXT_CONTRADICTION: the image content contradicts the slide text.
Report a finding only when you are confident the defect is actually present.
Output ONLY one JSON object (no prose, no code fences) matching:
{“has_defect”:  <bool>, “findings”:  [{“type”:  “<one candidate ID>”, “severity”:
“minor|moderate|severe”, “locator”:  {“level”:  “page”, “page_id”:  “<id>”, “element_id”:
```

```
“<id-or-null>”, “related_page_ids”: []}, “evidence”: “<short grounded reason>”, “fix_suggestion”:
“<one executable action>”}]}
If the slide/deck is clean, output {“has_defect”: false, “findings”: []}.
```

C3 system

```
You are a meticulous slide-quality inspector. You will be asked about ONE specific possible defect.
Look only at what is visibly rendered. Answer with a single JSON object and nothing else.
```

C3 user

```
[IMAGE]
Question: [one question from the list below]
Answer strictly as JSON:
{“present”: true|false, “evidence_element”: “<concrete element/region you point to, or empty>”,
“evidence_region”: “<top-left|top|top-right|left|
center|right|bottom-left|bottom|bottom-right|empty>”, “confidence”: 0.0-1.0}
If present=true you MUST name a concrete evidence_element (e.g. “the title”, “the right-hand
card”, “the bottom list item”). If you cannot point to concrete visible evidence, answer
present=false.
```

The instantiated C3 question for each evaluated class was:

- G1: Does any text visibly overflow or get clipped by its text box—letters cut off at an edge, or a line running past the box boundary?
- G2: Does this slide visibly exhibit the following defect—Two non-background elements overlap.?
- G3: Among a list of bullet/body items that should line up, is ONE of them misaligned—indented or shifted so it does not line up with the rest?
- G4: Does this slide visibly exhibit the following defect—Same-level text uses inconsistent font sizes.?
- G5: Among a list of bullet/body items that share one text colour, does ONE of them have a text colour that clearly differs from the others?
- G6: Is the slide's whole block of content shifted toward one side, leaving clearly unequal left/right margins—one side crowded against (or running off) the edge while the opposite side is noticeably empty?
- S4: Does this slide visibly exhibit the following defect—Text density exceeds a scenario-specific rule.?

2 Coverage precision companion

Table 3 is the precision companion to the main paper's balanced-accuracy coverage table, using the same E8-corrected corpora, model, named-attribution rule, and routing. "Linter+C3" is the configured per-class composition (linter on linter-owned classes, C3 otherwise), not an unscored OR assembled from marginal rates. A displayed zero includes cells in which the scorer made no positive predictions, for which the analysis code's declared point-estimate convention is precision zero.

3 Full multi-reward / VLM-judge audit grid

Table 4 expands the main paper's Table 3 (which reports only G5 and G7) to the full per-class paired preference accuracy $P(r_{\text{clean}} > r_{\text{def}})$ for all audited classes, with 95% CIs on G7. Each scorer penalizes several in-taxonomy defects it can see, confirming each is a functioning scorer rather than a domain mismatch; the G7 column is the linter-blind render class whose detection splits the scorers by perceptual capability, not training domain (§5.3 of the main paper). The full multiplicity report is included in the Code and Data Supplement (`reports/`).

Table 3: Per-cell precision corresponding exactly to the main paper's coverage table. Values in brackets are 95% Wilson intervals when defined.

class	linter	VLM-C0	VLM-C3	linter+C3	routed
G1	1.00 [.91,1]	0	0	1.00 [.91,1]	1.00 [.91,1]
G2	1.00 [.86,1]	1.00 [.82,1]	1.00 [.88,1]	1.00 [.86,1]	1.00 [.86,1]
G3	1.00 [.94,1]	1.00 [.82,1]	1.00 [.88,1]	1.00 [.94,1]	1.00 [.94,1]
G5	1.00 [.91,1]	0	1.00 [.79,1]	1.00 [.91,1]	1.00 [.91,1]
G6	1.00 [.91,1]	0	.506 [.429,.584]	1.00 [.91,1]	1.00 [.91,1]
G7	0	0	1.00 [.91,1]	1.00 [.91,1]	1.00 [.91,1]
S1	0	.900 [.699,.972]	.750 [.551,.880]	.750 [.551,.880]	.900 [.699,.972]
S4	0	.667 [.208,.939]	1.00 [.89,1]	1.00 [.89,1]	1.00 [.806,1]
S6	0	.500 [.314,.686]	.500 [.314,.686]	.500 [.314,.686]	.500 [.314,.686]

Table 4: Multi-reward / VLM-judge audit: paired preference accuracy per class (chance = 0.5; G7 with 95% CI). ✓/× marks reliable G7 detection (CI above / spanning 0.5). G5 is the internal-chromatic re-operationalisation of §4, a *visible* defect most scorers catch, not the linter-only blind spot G7 is.

Reward scorer	category	G1	G2	G5	**G7**	S4	S6
DocReward-3B	document	0.93	1.00	0.72	× **0.48** [.38, .58]	1.00	1.00
Skywork-VL-7B	general-mm	0.94	0.72	0.57	✓ **0.79** [.69, .86]	0.92	1.00
PickScore-v1	general-mm	0.74	0.59	0.70	✓ **0.71** [.61, .80]	1.00	0.72
LAION-Aesth.	aesthetic	0.37	0.91	0.78	× **0.57** [.46, .66]	0.75	0.61
CLIP-IQA	aesthetic	0.44	0.50	0.65	✓ **0.83** [.74, .90]	0.50	0.72
Qwen3.7-Max judge	frontier VLM	0.13	0.92	–	✓ **1.00** [.86, 1]	1.00	–
n (paired; non-VLM rows)		54	54	80	90	36	18
n (Qwen3.7-Max judge row)		24	24	–	24	24	–

4 Full page-semantic examiner table

Table 5 reproduces the page-semantic inspection table of §6 of the main paper, here shown with the deck-scope S2/S5 negative result and the full per-cell context.

5 Human spot-check v2

The rebuilt-corpus human spot-check used in the revised limitations text is summarised in Table 6. To avoid rewriting the full 69-pair sample for a definition bug confined to legacy G3/G5, we kept the other 55 pairs fixed and replaced only those stale samples with corrected internal- contrast variants (within-list misalignment for G3, within-list chromatic clash for G5). The resulting v2 sample has 73 pairs. The paired clean controls remain clean throughout, and the source-structure faithfulness audit on the auditable classes is 55/55 (see `data/part3/e8_ir_faithfulness_v2.json` in the Code and Data Supplement). A short delta summary is provided in `reports/_e8_spotcheck_v2_delta.md`.

6 Extended figures

These figures were omitted from the main paper for space; their findings are stated in the body prose. They are included here for completeness.

Table 5: Page-semantic inspection, balanced accuracy [95% Wilson] on paired clean (best modality). The finetuned 8B examiner beats both zero-shot baselines, including the 30B, *in-distribution*. Deck-scope S2/S5 are a negative result. n = paired positives (2-AFC rows = forced-choice pairs); the S6 2-AFC rests on only 12 pairs and is descriptive only.

class	**ft-8B**	zs-8B	zs-30B	n
S1 title/body	**1.000** [.95,1]	0.778 [.69,.85]	0.933 [.87,.95]	36
S4 density	**1.000** [.97,1]	0.529 [.50,.58]	0.774 [.69,.84]	60
G1 overflow (2-AFC)	**1.000** [.97,1]	0.975 [.93,.99]	0.700 [.61,.77]	120
S6 image/text (2-AFC)	**1.000** [.76,1]	1.000 [.76,1]	0.500 [.25,.75]	12
S2 narrative (deck)	0.500† [.45,.53]	0.646 [.53,.72]	0.549 [.42,.66]	36
S5 missing-logic (deck)	0.500 [.47,.53]	0.500 [.47,.53]	0.567 [.45,.68]	60

† degenerate (always-flag): the SFT mix had no clean-*deck* negatives. See §6 of the main paper.

Table 6: Human spot-check v2 on the rebuilt corpus. Rates are human perceptual-verification rates on the defective render and clean-twin verification rates on the paired clean render.

class	defect visible	twin clean	n
G1	1.00 [.70,1]	1.00 [.70,1]	9
G2	1.00 [.65,1]	1.00 [.65,1]	7
G3	1.00 [.68,1]	1.00 [.68,1]	8
G5	1.00 [.72,1]	1.00 [.72,1]	10
G6	1.00 [.65,1]	1.00 [.65,1]	7
G7	1.00 [.70,1]	1.00 [.70,1]	9
S1	1.00 [.65,1]	1.00 [.65,1]	7
S4	1.00 [.65,1]	1.00 [.65,1]	7
S6	1.00 [.70,1]	1.00 [.70,1]	9
overall	**1.00 [.95,1]**	**1.00 [.95,1]**	**73**

7 Frozen-route held-out and extreme-offset addenda

Independent held-out protocol. We generated a seed-controlled calibration corpus from templates not used by the development experiments: a coloured header, accent rail, footer, and visible diagram are varied across eight layouts and six topics. Each of nine classes has 150 defective slides and 150 same-base clean controls. Every labelled sample has a unique clean twin; all 1,350 pairs differ in rendered pixels. The fidelity gate found all 750 declared-geometry positives and zero target or incidental linter findings on their clean controls. For G7, all 150 declared IRs are linter-blind and all 150 rendered overflow pairs pass the localization check. We then evaluated the corrected frozen route exactly once: G1/G2/G3/G5/G6→linter, G7→VLM-C3, S1/S6→VLM-C0, and S4→text LLM. Table 7 reports named attribution; all cells use $n_+ = n_- = 150$.

The held-out result preserves the routing conclusion but also exposes a specific limitation rather than hiding it in a mean: S6 has perfect recall but 97/150 clean false positives, so its C0 route fails the strict precision criterion on the novel diagram family. The corrected S1 visual route, by contrast, transfers without an error.

Extreme G6 magnitude boundary. To stress the moderate-offset result without changing its operationalization, we rigidly translated the whole content group left by 144 px, preserving all internal alignments and placing its left edge exactly 48 px beyond the canvas. On 12 positive and 12 paired-clean slides, both pointwise C0 and atomic C3 reach .958 [.702,.993] detection balanced accuracy (11/12 positives, 12/12 clean; precision 1.000 [.741,1]). Their attribution differs: C0 detects generic damage but names G6 on 0/12, whereas C3 names it on 11/12. Therefore the earlier failure is a moderate-magnitude global-position blind

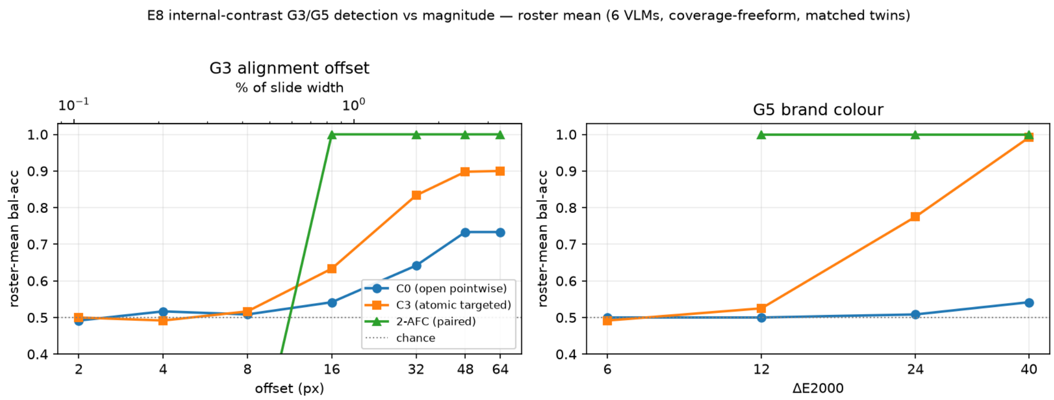

Figure 1: Fine geometry is magnitude-gated, not a flat capability floor (referenced as the saturation curve in §4). Roster-mean balanced accuracy (6 VLMs, matched-twin internal-contrast set) vs. injected magnitude for G3 alignment (left) and G5 brand color (right). Open pointwise (C0) lags; atomic targeted (C3) recovers to ≈ 0.90 by 48–64 px / 0.99 by ΔE 40; the paired 2-AFC saturates to 1.00 by 16 px (0.83%) / ΔE 12. The sub-threshold tail (≤ 8 px, ≤ 6 ΔE) stays at chance under every protocol.

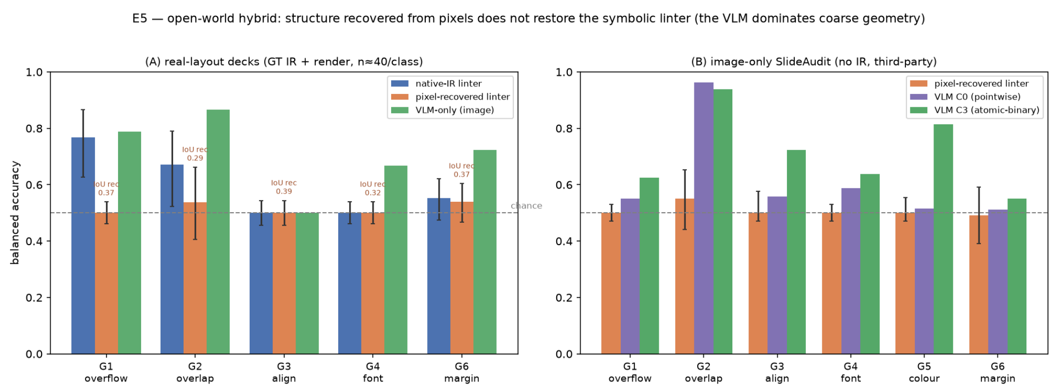

Figure 2: Open-world hybrid: structure recovered from pixels does not restore the symbolic linter (referenced in §7). **(A)** real-layout decks (GT IR + render, n≈40/class): native-IR linter vs. pixel-recovered linter vs. VLM-only floor, Wilson CIs; recovered structure rescues only overlap (G2) and is silent on fine geometry, while the VLM dominates every class. **(B)** image-only SlideAudit (no IR, third-party): the recovered linter clears chance only on G2; the C3-elicited VLM is far stronger.

spot, not a claim that a VLM remains blind after the page visibly clips content.

8 Conceptual figures

These vector "concept" figures (with text overlays) were used in the single-column technical report and are reproduced here for readers who prefer the pictorial version of the protocol and routing arguments. They convey no information beyond the body prose and the data figures above; they are included for readability.

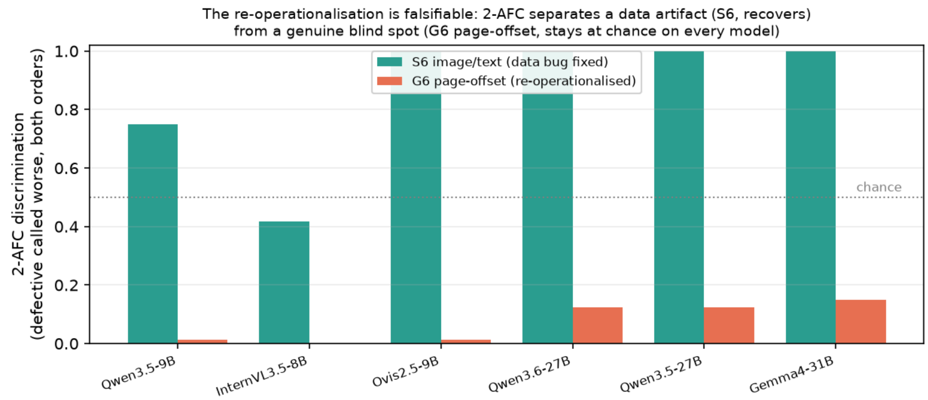

Figure 3: A magnitude-bounded blind spot the forced choice exposes (referenced in §4). Per-model 2-AFC discrimination. Pointwise puts both classes at chance, but the forced choice *separates* them: the data artifact S6 recovers on every model, while G6 page offset stays at the floor on every model at the tested moderate, unclipped magnitudes. A separate extreme-offset stress test below locates the clipping boundary where this conclusion ceases to hold.

Table 7: One-shot corrected frozen-route evaluation on the independent held-out corpus. Intervals are 95% Wilson intervals. The predeclared covered criterion is balanced accuracy and precision both ≥ .70.

Class (route)	Balanced accuracy	Precision	Covered
G1 (linter)	1.000 [.975,1]	1.000 [.975,1]	yes
G2 (linter)	1.000 [.975,1]	1.000 [.975,1]	yes
G3 (linter)	1.000 [.975,1]	1.000 [.975,1]	yes
G5 (linter)	1.000 [.975,1]	1.000 [.975,1]	yes
G6 (linter)	1.000 [.975,1]	1.000 [.975,1]	yes
G7 (VLM-C3)	.937 [.886,.965]	.917 [.863,.951]	yes
S1 (VLM-C0)	1.000 [.975,1]	1.000 [.975,1]	yes
S4 (text LLM)	1.000 [.975,1]	1.000 [.975,1]	yes
S6 (VLM-C0)	.677 [.628,.716]	.607 [.545,.666]	no
Macro mean	.957	—	8/9

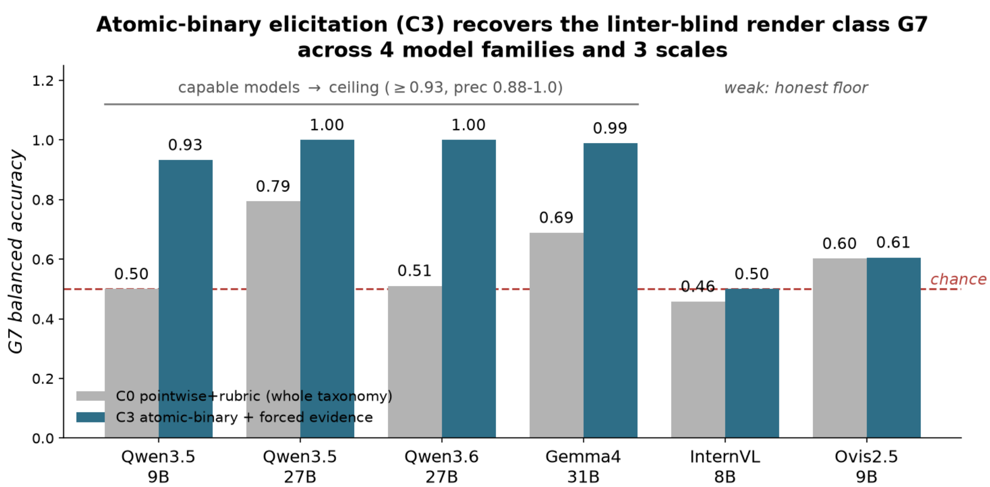

Figure 4: Atomic-binary elicitation (C3) recovers the linter-blind render class G7 across the capable subset (referenced in §5.1). Capable models reach the ceiling (≥ 0.93, precision 0.88–1.0); the weaker InternVL/Ovis stay near their perceptual floor, a capability-dependent boundary, not a universal fix.

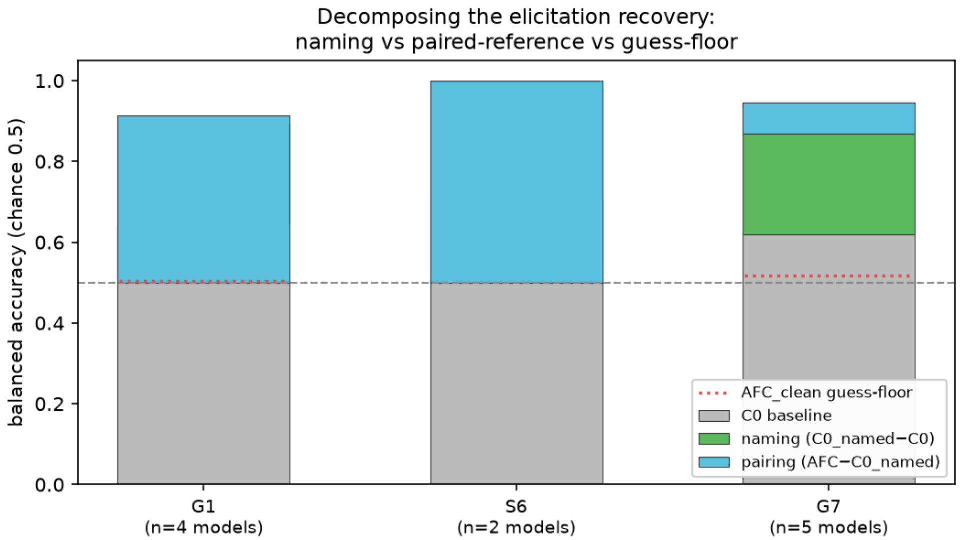

Figure 5: What drives the recovery: naming vs. a paired reference (referenced in §5.1). The chance→ceiling jump factors into a *naming* component (C0_named−C0) and a *pairing* component (AFC−C0_named). For G7 the naming term carries it (green); for G1 and S6 the paired reference does (blue). The AFC_clean guess-floor (dotted) is near chance, so neither is a forced-pick artifact.

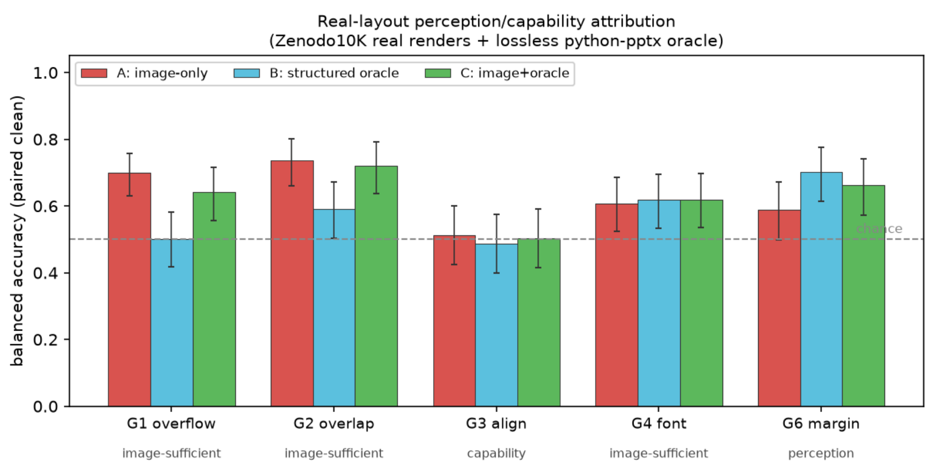

Figure 6: The perception/capability split reproduces on real layouts (referenced in §7). ZENODO10K real renders + lossless `python-pptx` oracle; mean over 3 VLMs / 3 families, balanced accuracy on paired clean; 95% Wilson on counts pooled across models (n=76–90 per class). Coarse defects are image-sufficient; G6 margin is a perception bottleneck the oracle rescues (B>A); G3 alignment stays near chance in every channel on real cluttered decks.

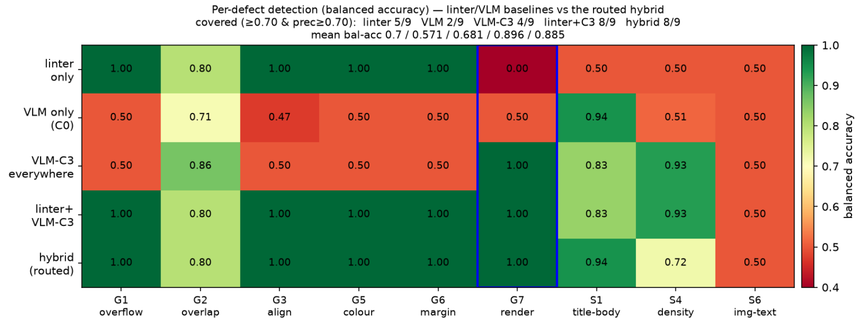

Figure 7: Coverage by engine (companion to Table 2 of the main paper). Per-defect balanced accuracy for the linter, VLM-C0, VLM-C3, linter+VLM-C3, and the routed hybrid. Minimal linter⊕C3 has the highest mean balanced accuracy; G7 (boxed) is the linter-blind cell rescued by C3.

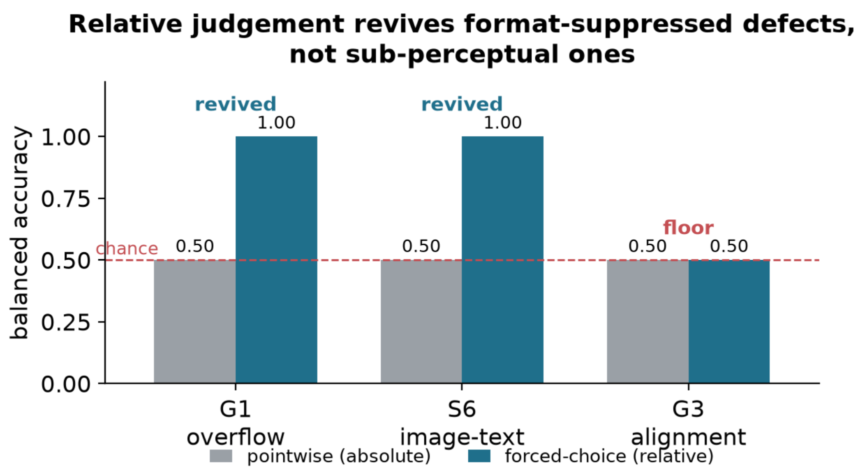

Figure 8: Relative judgement recovers absolute-pointwise blind spots (referenced in §4). For G1 overflow and S6 image/text contradiction, pointwise (absolute) inspection is at chance but a forced choice recovers perfect detection. G3 alignment behaves the same way above a magnitude threshold; only its sub-threshold tail and S3 terminology stay at chance side by side.

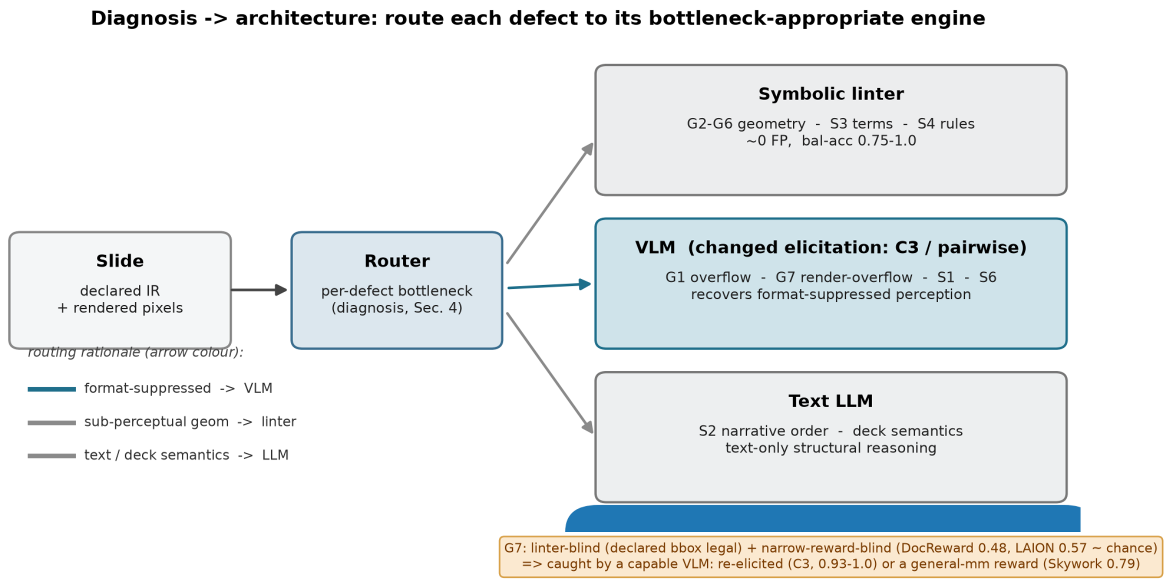

Figure 9: Diagnosis → architecture. Route each defect to the engine that owns its bottleneck: declared geometry to a symbolic linter, format-suppressed perceivable classes to a re-elicited VLM, text/deck semantics to an LLM. The G7 callout marks the linter-blind render-overflow class caught only by an engine with a rendered-quality read-out (Sec. 5 of the main paper). Companion to the teaser (Fig. 1).

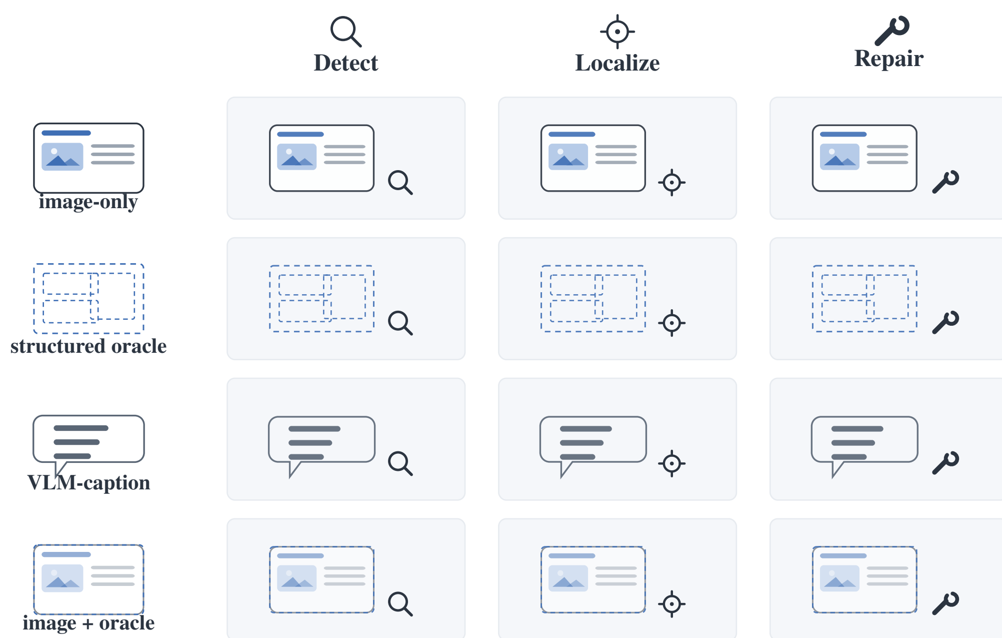

Figure 10: Attribution protocol. Each paired (defective, clean) slide is presented under four input modalities (image-only, the lossless structured oracle, a fixed VLM-caption control, and image+oracle) crossed with detect/localize/repair. Comparing image-only against the structured oracle isolates a clean perception/reasoning split (Sec. 3 of the main paper).

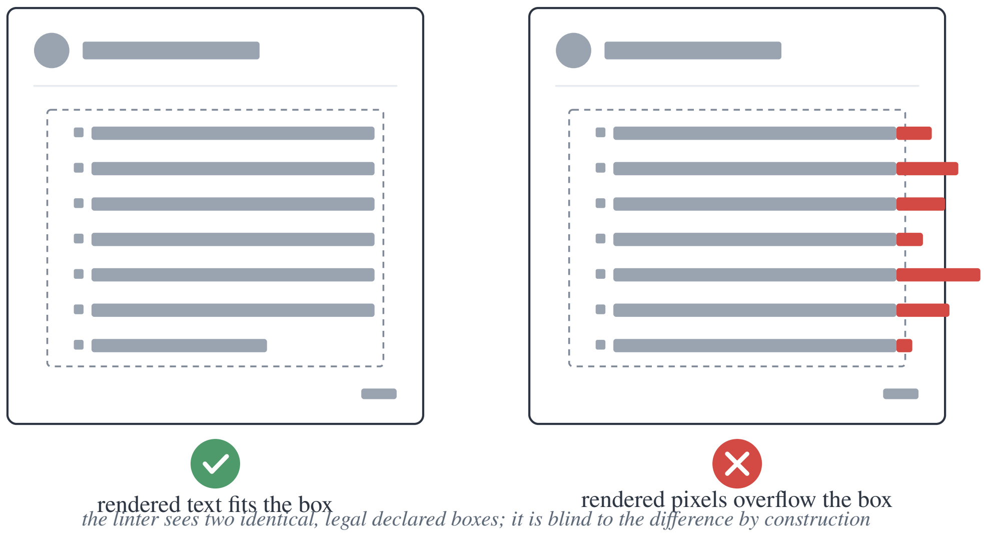

Figure 11: G7 render-overflow. The declared box (dashed) is legal and *identical* in both slides; only the rendered pixels overflow on the right. The linter and any structure-only model read the same legal box in both, so they are blind to G7 by construction (Sec. 5.3 of the main paper).

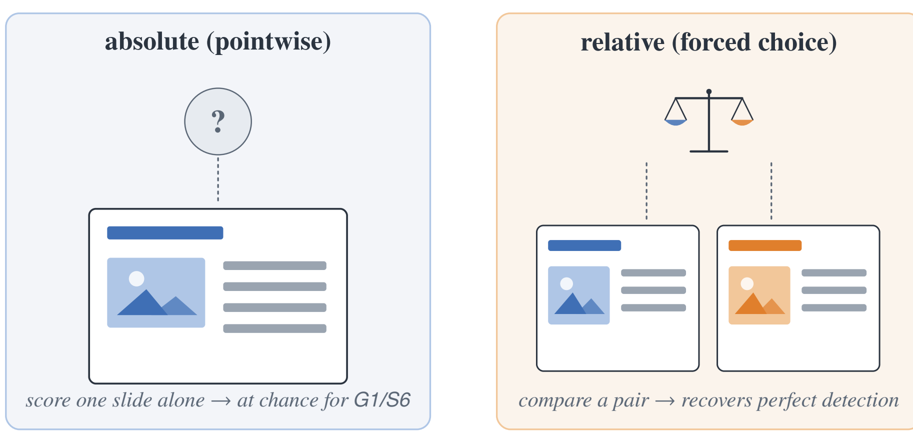

Figure 12: Two elicitations, same model and images. Pointwise (absolute) inspects one slide in isolation; the forced choice compares a pair. For G1 and S6, changing absolute to relative is what recovers detection (Sec. 4 of the main paper).

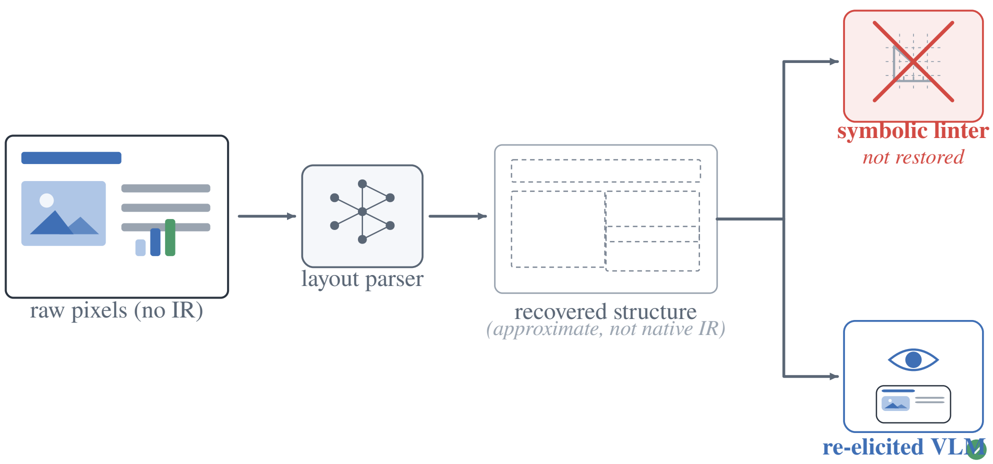

Figure 13: Open world: recovered structure does not restore the linter. For bare third-party pixels with no IR, a layout parser recovers approximate structure, but it is not the native IR the symbolic linter needs; only the re-elicited VLM remains a deployable critic (Sec. 7 of the main paper).